Samuel Tribble

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EDUCATION

Harvey Mudd College **Claremont, CA**

B. S. Computer Science *Class of 2028*

- Coursework: CSCI035: Computer Science for Insight (2024), PHIL179E: Philosophy of Artificial Intelligence (2024), CSCI060: Principles of Computer Science (2024), CSCI070: Data Structures / Program Development (2025), ENGR191: Advanced Problems in Engineering (2024-2025).
- Awards: Cobell Undergraduate Scholarship recipient for the 2024-2025 and 2025-2026 academic years.

COMPUTER SKILLS

Python, C++, MATLAB, Java, SQL, Ubuntu (20.04), ROS 2 (foxy/humble), Gazebo, RViz2, VS Code, Git/GitHub, Claude Sonnet 5.5, hardware design with Raspberry Pi microcontrollers and Radio-frequency (RF) sensors, PuTTY, Blender, Unreal Engine, Jupyter Notebook, Excel.

COMPUTER WORK

Mars Sandstorm Box (3D printing + RC components + RF sensors + Raspberry Pi + PuTTY)

- Created hardware to simulate a physical Martian sandstorm and generate noisy localization measurements from Ultrawide-Band (UWB) sensors.
- Wrote software to interact with data output from a network of Ultrawide-Band (UWB) sensors via PuTTY terminals.
- Used a Raspberry Pi, electronic speed control (ESC), and Python to remote-control a fan for the sandstorm.

Pilgrim Rover LiDAR Noise Injection (ROS 2 + Gazebo)

- Programmed a ROS 2 Python node that subscribes to a ROS 2 topic node, /scan, from a TurtleBot3 burger, injects Gaussian noise, and publishes the output to a ROS 2 topic node, /scan_noisy.
- Wrote data analysis software with Python, NumPy, Pandas, and Matplotlib to generate a histogram of rover LiDAR data sampled in normal weather conditions vs hazardous weather conditions.

ECHO - A 3D Autonomous Rover Simulator (ROS 2 + Unreal Engine)

- Leveraged existing high-fidelity 3D models of Martian terrain and a rover that accepts manual input to design a prototype test bed.
- Extended the project base with C++ and Python to integrate ROS 2 and vision tools (e.g., YOLO).

Insight - A Machine-learning Framework for Bounding Localization Error

- Using Pandas, NumPy, Scikit-learn, and Matplotlib, I co-authored Python software to implement conformal prediction algorithms that bound the localization error in noisy position measurements of an autonomous rover (i.e., TurtleBot3).
- Extended the project base to financial data and generated over $10,000 in value through trading decisions aided by algorithmic stock price forecasts.

PROFESSIONAL EXPERIENCE

Harvey Mudd College **Claremont, CA**

Undergraduate Research – Research Fellow and Mentor Aug. 2024 – Nov. 2026

- Developing a digital test bed with the Unreal Engine and ROS 2 to simulate autonomous Lunar/Martian rover tasks and evaluate vision models (e.g., YOLO for object detection on a 3D model of a radio game asset).
- Researching radio-frequency (RF) Ultra-wide Band (UWB) sensors and engineering hardware simulations of UWB networks in degraded Lunar/Martian environments (e.g., rover localization with a JetBot-mini or TurtleBot3 driving along a planned route in a sandstorm).
- Co-authored an academic paper on the development of large-language models (LLMs) for generating synthetic datasets of Lunar/Martian terrain classes.
- Mentored students in undergraduate research and developing deliverables in an engineering context.
- Co-authored localization algorithms with Python, NumPy, Pandas, Matplotlib and scikit-learn machine learning tools that bound/quantify the uncertainty in output from Ultra-wide Band (UWB) sensors.

INTERESTS

Aerospace, robotics, game modding, Anishinaabe philosophy, skateboarding, rock music, chocolatiering, DUCK! (a captain of the HMC improv club), investing/trading on the U.S. stock market.